

Your second PCB portfolio piece

Upwork demand review + STM32 Industrial I/O scope assessment

Prepared October 4, 2026 | Ray Malik / MuffinByteLabs

Recommendation: follow through, with a firm scope limit

Yes. The controller is a credible second portfolio project for embedded, mixed-signal and industrial-control work. Its functions belong together: receive sensor signals, switch low-voltage loads, report data over wired networks, and recover predictably from faults. The most valuable next addition is finished firmware and measured evidence, rather than another major hardware block.

The project is already ambitious. Freeze the major functions, complete a useful Modbus-controlled board, then qualify classic CAN and the remaining performance envelope. Add wireless or Ethernet only if you deliberately choose a connected-controller application after the base system works.

What was reviewed	Result
Supplied Upwork sample	7 pasted files; 350 cards; 349 distinct listing texts; 347 deduplicated job briefs after two clear repost merges.
Actual selected controller	STM32G474VET6, integrated directly on a four-layer PCB. The ESP32S3 folder name is stale relative to the active project documentation.
Current maturity	Architecture, exact circuit/support selections, calculations and test plans. Native schematic, PCB layout, application firmware and prototype measurements are still pending.
Strongest market overlap	Complete PCB/embedded delivery; power conversion and protection; MCU/USB; analog acquisition; multilayer layout; CAN; test and manufacturing handoff.
Main uncovered market segment	ESP32 / wireless / battery-powered products. This is a real portfolio gap only to the extent your earlier plant-monitor board does not already demonstrate it.
Largest practical risk	Complexity delaying the first working, documented prototype. Adding channels or niche interfaces has limited value compared with completion.

Evidence distinction: listing counts show explicit topics in the supplied sample; project coverage is planned; suitability and priority are engineering judgments. This report does not establish electrical correctness or fabrication readiness.

How the sample was evaluated

Every listing description was extracted from all seven supplied files. Search filters, skill badges and footer text were excluded from feature counting. Each deduplicated brief counts once per topic, even if the term appears repeatedly. One exact duplicate and two clear underlying-project repost pairs were merged; generic repeated titles with different descriptions or distinct paid scopes were retained. This is not a claim of 347 distinct clients or products.

Duplicate / repost handling	Decision
J024: PCBA Routing (and layout)	Exact repeat in F1 card 24 and F3 card 7; one project.
J029 + J042: wearable EEG miniaturization	Same underlying wearable project; descriptions retained as variants.
J060 + J068: Adafruit FX 2217 power-down	Same audio power-down project; corrected follow-up version retained.
PCB Design / PCB Designer / PCB Design Engineer	Identical generic titles with different scopes; kept as separate projects.

What a count means

The tables count **explicit topic references in titles or descriptions**. A reference can describe required hardware, desired experience, an optional scope, a review target or a deliverable. It is not automatically a mandatory onboard feature. Broad categories are labeled to include their actual search terms. Manually reviewed exclusions and evidence snippets are recorded in the companion data.

A low count does not mean a skill has no market. Short descriptions can omit details; some requests offer alternative platforms; some are firmware-only, consulting or repair rather than new PCB design. The sample is selected by a broad search, contains noisy matches and is not a random or complete census of Upwork.

The decision rule used for your board

- Repeated in more than two independent projects: consider it, then check application relevance and implementation cost.
- Only one or two projects: do not treat it as a required addition. A coherent feature already selected can remain for its application value.
- Broad service requirements: demonstrate them through editable design files, firmware, layout reasoning, bench results and handoff documents.
- Platform-specific or different-domain requests: acknowledge the gap without claiming the I/O controller proves that expertise.

The pasted listings contain relative posting ages but no reliable capture date or original job URLs. They are treated as the supplied historical sample, not verified current vacancies. Additional browsing was limited to primary technical and portfolio guidance; web results were not added to the sample counts.

Recurring topics your plan can demonstrate

Counts below use 347 deduplicated job briefs. These are positive topic references, including desired experience or optional work, not a percentage of jobs this board will qualify you for. Categories overlap and must not be added together.

Topic in supplied listings	Briefs	Your planned demonstration
Gerbers / editable native design	77 / 70	Release-quality native CAD and coherent fabrication/assembly package.
Firmware / embedded software	94	STM32 source, documented builds and tested I/O/protocol/fault behavior.
Bench testing / bring-up	53	Physical assembly, measured results and reproducible fault/communication tests.
Power conversion / power supplies / management	76	9-30 V input, buck/LDO conversion, budgets and measured rail behavior.
Non-audio analog sensing / acquisition	22	Manually reviewed sensing/front-end subset. Protected acquisition, filtering, calibration and error budget; not an exact industrial-format count.
Protection / ESD / surge topics	47	Defined supply/terminal fault handling and layout. Component selections are not immunity test results.
USB	53	USB-C full-speed data service. Does not demonstrate USB high-speed, host capability or PD charging.
Four-layer PCB	29	Ground/reference strategy, mixed-signal placement, switching loops and isolation boundaries.
EMI / EMC topics	27	Explain layout choices and collect appropriate test evidence. This is not a certification claim.
STM32	26	Directly integrated MCU, debug, clock/reset/power design and firmware.
ADC specifically	21	External ADC communication and measured system accuracy, not merely resolution.
I2C / SPI / UART	23 / 18 / 12	Already used by EEPROM, external ADC and RS-485. Extra breakout connectors are not necessary to prove these buses.
Relays / PWM	14 / 6	Dry contacts and a bounded high-side PWM channel. These counts do not establish demand for your exact quantity or rating.
CAN / RS-485	12 / 6	Meaningful repeated wired-interface coverage, with classic CAN brought up before FD.

The strongest conclusion is broad transferable skill alignment. Industrial bus and exact sensor-format demand is a smaller subset of the sample. The board is a strong focused example, not a complete response to the entire search.

Current functions: keep them for a clear purpose

Selected function	Fit and recommended treatment
STM32G474 + four layers	Keep. Shows a different MCU/design style if your first board used a wireless module. No platform change is justified purely by ESP32 keyword frequency.
Protected nominal 12/24 V supply	Keep. Appropriate to industrial sensors/actuators and transferable power/protection work. 9-30 V is a target pending qualification.
4 group-isolated DI + pulse counting	Keep. Coherent sensor/contact acquisition. DI_COM is one isolated group; channels are not individually isolated. Inputs accept positive signals from PNP/sourcing sensors; NPN/AC signals need a different reviewed interface.
2 x 0-10 V + 2 x 4-20 mA inputs	Keep for industrial measurement, while recognizing these exact formats are not repeatedly requested in this sample. If reducing cost, fewer populated channels can prove the same engineering skill.
External 16-bit ADC + calibration	Keep if the precision-acquisition story is a target. Demonstrate loading/noise/drift/settling and independent verification. Resolution is not an accuracy result.
4 diagnosed high-side 0.5 A outputs	Keep. Actuator switching plus current/fault telemetry adds useful breadth beyond sensing. Ratings and combined loading remain targets.
DO3 bounded 100 Hz PWM	Keep the available mode, qualify it after on/off behavior. It does not cover BLDC, FOC, H-bridges, arbitrary fans or proportional solenoids.
2 SPDT dry-contact relays	Coherent, but the second relay offers little new portfolio skill. Populate one first or reduce to one if size/power/cost needs it. Contacts remain low-voltage DC.
Isolated RS-485 / Modbus RTU	Keep as the first complete command/measurement interface. Modbus references in the sample are mostly TCP/system integration, so RTU is an application choice rather than a demonstrated frequent exact request.
Independent isolated CAN / CAN FD	Keep CAN. Classic CAN provides the principal initial proof; defer FD testing until the base board works. Isolation is a deliberate robustness choice, not required by every CAN client.
USB-C, SWD/SWO, EEPROM, watchdog/ARM	Keep. Service, recoverable settings, debugging and default-off behavior support a complete usable product.

Project basis: README.md; docs/Architecture.md; docs/Interfaces.md; docs/circuits/Control_Service.md, Analog.md and Field_IO.md; firmware/README.md. All functional coverage above is planned, not implemented.

Important precision: exact interfaces vs demand

4-20 mA is useful, but not a repeated explicit request here

There are 0 literal 4-20 mA references in the supplied sample. That does not invalidate the interface: it is appropriate to industrial signal acquisition, and TI's PLC input reference design uses both current and voltage sensor ranges [S1]. It means the decision is supported by application coherence and transferable analog skills, rather than a recurring exact Upwork request.

0-10 V input and output are different skills

The 2 exact 0-10 V references are J198 (lighting control/dimming) and J271 (off-the-shelf HVAC EC-fan controller advice). They concern output/control applications, not your planned voltage-sensing inputs. Under your one-or-two-job rule they do not justify adding an analog output to Rev A.

CAN FD controller references are not always FD traffic requests

J045 selects an MCU with FDCAN but explicitly asks for classic CAN at 500 kbit/s. A term count that includes FDCAN must not be described as a count of mandatory FD applications. Your STM32G474 has an FD-capable peripheral [S2]; demonstrating classic CAN first is still directly useful.

Modbus RTU is a good application choice, with modest direct sample support

There are 6 RS-485 and 3 Modbus topic references. The Modbus examples are largely house/PLC/prop integration and Modbus TCP. RS-485 hardware demand is clearer in J003, J019, J077, J164 and J246. Use RTU because it makes this controller practical; do not call it one of the dominant exact requests in the sample.

Your existing buses already cover common MCU interfaces

The SPI ADC, I2C configuration EEPROM and UART-based RS-485 are purposeful implementations. Document and test them. A spare header alone would not prove those skills, and an external wireless gateway would demonstrate integration rather than onboard RF layout.

Keep the product boundaries visible

Analog inputs share MAIN_GND and are not galvanically isolated. Current loops are externally powered and have a planned 4.25 V burden at 20 mA before wiring. High-side current telemetry is multiplexed diagnostic data. The 100 Hz publication rate does not establish 100 Hz input bandwidth; the analog plan targets useful input bandwidth up to 20 Hz. These boundaries help clients judge relevance accurately.

Missing market areas: which ones belong here?

Uncovered topic	Assessment	Recommended action
ESP32 (54); BLE/Bluetooth (45); Wi-Fi (25)	Substantial repeated demand. Wireless commissioning and remote telemetry can make sense on an I/O controller. Wireless wearable experience remains a distinct specialization.	Keep STM32 for Rev A. If your plant monitor lacks wireless proof, a bounded module/gateway extension is the strongest later option. Do not add two MCU firmware stacks just for keywords.
Ethernet / PoE (17)	Ethernet can suit remote industrial I/O. Count includes Ethernet switches, carrier boards, PLC integration and PoE; those are different scopes.	Optional Rev B/network expansion after a working base. STM32G474 has no native Ethernet MAC; a simple PHY alone will not add Ethernet [S2]. Adding an external controller also requires firmware/protection/layout work.
Battery (75); charging (41); low-power topics (40)	Common in the search, mainly different product/power objectives. A continuously powered 24 V actuator controller is a poor showcase for wearable sleep current.	Do not add charger/BMS/backup battery by default. Use a separate low-power project or your existing monitor if it actually demonstrates it.
Analog command output / DAC	Some DAC mentions are generic peripheral experience, audio or different power-control applications. Exact 0-10 V output relevance is only two projects.	Do not add now. A future actual valve/drive application could justify it, but PWM load pulses are not a precision analog command.
H-bridge / motor driver / FOC	Motor topics recur, but require a different energy, switching, sensing and control design than the selected high-side output.	Do not add a motor stage to cover that segment. Present bounded on/off inductive-load control accurately.
Display, audio, GPS/cellular, cameras, UWB, FPGA/DDR	Some are frequent, some rare. Most change the product objective or require distinct specialist proof.	Exclude from this controller unless a real application requires them. A PC service tool provides useful visibility without onboard HMI hardware.

No forced random feature additions are recommended. If one later hardware expansion is selected, choose one connected-I/O use case and prove it thoroughly. A radio module needs correct placement/antenna clearance, supply integrity and actual link/firmware tests; a connector or gateway does not prove that onboard layout work.

What to add now: usable software and evidence

1. A small USB diagnostic and calibration tool

USB CDC is already planned; a usable host tool does not exist yet. Add live calibrated inputs with units, raw values, timestamps and validity; explicit arm/disarm and output commands; current/fault status; coefficient editing with validation; and recorded test-data export. A straightforward command-line tool or small desktop panel is enough. Use it for service and bench evidence rather than adding a screen to the PCB.

2. A frozen data contract and reproducible firmware

Define Modbus register addresses, units, signedness, word/byte order, atomic snapshots, validity, exceptions and command-leak behavior. Define CAN IDs and payload versions when CAN is implemented. Include exact toolchain versions, build/programming instructions, a recoverable service path and firmware/build identity in logs. RTOS use should follow a scheduling need; a deterministic scheduler is already a valid choice.

3. One complete application demonstration

Use a small industrial bench fixture: a pulse/contact sensor and voltage/current input represent machine state; a 24 V lamp or bounded coil represents an actuator; a Modbus host commands the controller and logs calibrated feedback. Show a command timeout or reset during an active output, hardware inhibition, fault reporting and fresh rearming. Add a CAN peer after the first complete demonstration works.

4. Evidence clients can review

- Board photos and a short demonstration video tied to hardware/firmware revision.
- Calibration residual plots at independent points, with reference uncertainty and switching conditions.
- Supply/output/PWM/reset captures showing why layout and fault-handling decisions matter.
- Thermal measurements under stated combined loading and enclosure/ambient conditions.
- A coherent editable CAD, manufacturing, mechanical and firmware release package.

These are completion deliverables, not absent hardware blocks. Your repository already plans much of this; implementing and publishing it is the missing proof. Upwork guidance also favors relevant personal projects with clear outcomes and supporting visuals [S4].

What to simplify or defer

Do not increase channel count for market coverage

Four versus eight similar channels rarely doubles the relevant skill demonstrated. Keep the current channel counts if they serve the fixture and fit the budget; use staged population for bring-up. A second relay is an easy scope reduction if it competes with time, enclosure space or power, because one relay already proves dry-contact control.

Defer performance breadth before adding circuitry

Finish nominal 12/24 V, room-temperature acquisition and Modbus operation first. Bring up classic CAN next. Then close the full 9-30 V/0-50 C/combined-load targets, PWM timing/diagnostics and CAN FD. Deferred functions must remain labeled as pending; a subset of passed tests is not the full claimed rating.

The rail and protection architecture deserves a complexity review

The control selection allocates 11 dual-AND packages, 10 dual receiving buffers including loop-status qualification, four dual Schmitt packages, six supervisor/window-monitor packages, an external watchdog and an ARM latch. Separate same-reference logic supplies and wide powered-off/fault-survival requirements drive much of that support circuitry. This is a substantial second-project challenge. If the scope threatens completion, a narrower completed variant is a valid portfolio choice.

Consider a fresh architecture review of whether local MCU/field logic can share a digital supply while retaining the actual DI and bus isolation domains. Any consolidation must recheck noise, load capacity, startup/shutdown, powered-off injection and default-off behavior. The current buffers/gates serve real requirements; simply deleting them would break the rationale.

Likewise, the +15 V and threshold rails support the selected analog input protectors. They are not spare rails left over from an analog output. A simpler analog path would require a new protection/fault-envelope design and coordinated documentation, not just removing those converters.

Keep the first demonstration bounded

Do not expand PWM into a general motor driver, proportional-solenoid controller or arbitrary LED dimmer. Do not add universal NPN/AC input support, loop power, per-channel analog isolation or a mains supply solely to collect more feature names. The planned hardware watchdog and fault recovery are worthwhile, but do not represent a certified functional-safety system.

Open design gates already documented include CAD library completion, capacitor bias/hold-up evidence, hot input-path voltage-drop margin, analog error budget, output/PWM timing and enclosure thermal verification. Close those gates through implementation and measurement before increasing scope.

Listings that most clearly support this direction

J008 | PCB Layout for STM32G030F6P6 in KiCAD 10

Four-layer STM32 control/measurement board in KiCad. Explicit analog conditioning, a switching supply, grounding/power distribution, EMI/EMC-conscious layout, footprints and assembly handoff. This is a strong match to the skills your board can demonstrate. It explicitly excludes RF and BGA.

Source F1, card 8, lines 301-322

J045 | Hardware engineer: 10-channel current sensor PCB with CAN (STM32) - redesign + 4 prototypes

STM32 current measurement over CAN, 24 V protection, four-layer layout, calibration, switchable termination, DIN-rail integration and tested prototypes. Especially relevant evidence of mixed-signal/control work. Its selected sensor technology, exact MCU, nonisolated bus and classic CAN scope differ from your board.

Source F1, card 45, lines 1175-1200

J123 | Altium Schematic + 6-Layer PCB Design

28 V protection, several high-side switching channels, precision telemetry and a hardware interlock. Strong transferable power/control overlap; the full project also asks for Altium, six layers, RF and a compute module, so this controller would be supporting experience rather than a complete match.

Source F3, card 24, lines 642-663

J003 | Continuation of onngoing STM32 and ESP32 projects

Maintenance of STM32/ESP32 industrial IoT products: firmware, UART/RS-485, SPI, I2C, CAN, ADC/DAC, desktop diagnostics and structured handover. Supports the complete system/software story; also exposes your current MQTT/networking gap.

Source F1, card 3, lines 165-193

J164 | Complete PCB design (IoT environmental sensor reader) in KiCad, test prototypes, assemble units

KiCad sensor-reader completion, industrial RS-485, pulse counting and analog inputs, assembled/tested prototypes. Strong input/bring-up relevance; wireless, cellular and alternative power sources require separate proof.

Source F4, card 15, lines 431-455

Further evidence and deliberate non-matches

J077 | Senior Embedded Hardware & Firmware Engineer - PCB Design, Layout & ESP-IDF Development

A broad ESP32-S3 embedded project includes precision ADCs, CAN/RS-485, regulators, protection, debugging, test points, firmware and bring-up. These are useful transferable skills. Its wireless, Ethernet, battery, eight-layer and high-speed scope is substantially beyond this board; do not present your full-speed USB as high-speed proof.

Source F2, card 27, lines 730-754

J002 | 4-Layer PCB Layout & Routing - ESP32 IoT Controller

Four-layer ESP32 controller placement/routing asks for planes, return-current control, separation of power/sensor signals, DRC and manufacturing files. Your board covers the layout discipline; it does not demonstrate the required RF antenna placement.

Source F1, card 2, lines 137-164

J136 | PCB Layout Verification

Mixed-signal review asks for power sequencing, backpower, filtering, switching-noise coupling, USB, DFM and EMI/ESD. Your current architecture provides relevant case-study material. Its compact Bluetooth/audio product remains a different layout specialty.

Source F3, card 37, lines 936-957

J198 | PCB & Schematic Design for Zigbee-Based Commercial LED Lighting Controller

Commercial lighting controller uses Zigbee, PWM, DALI and 0-10 V dimming. It is one exact analog-command application; it does not justify adding the full lighting stack to your I/O board.

Source F4, card 49, lines 1211-1237

J271 | Condenser Control System for 2 Circuits

HVAC fan-control consultation uses a 0-10 V speed command and explicitly asks for an off-the-shelf controller rather than a custom PLC. It is not a custom PCB analog-input job.

Source F6, card 22, lines 641-654

Completion gates and portfolio claims

Gate	What makes it reviewable
A. Scope freeze	One application, accepted function limits and one coherent architecture. Label optional FD/PWM/wider qualification milestones. Resolve library/capacitor gates before capture freeze.
B. Native design	Complete editable KiCad schematic/PCB, datasheet pin/pad checks, ERC/DRC with explained exceptions, reviewed return paths/isolation/protection and enclosure fit.
C. Staged prototype	Power and MCU/service first; then one input/output path; then replicated channels, relays and communication. Record issues and corrections with revision identity.
D. Complete operating mode	A real Modbus host reads calibrated measurements, commands outputs and verifies fault/timeout/rearm behavior using buildable firmware.
E. Measurement closure	Use independent analog verification points, combined-load thermal checks, defined safe fault fixtures, bus recovery and the planned at-least-three-unit/24-hour test records where practical.
F. Portfolio release	Photos/video, concise application explanation, selected plots/captures, native design and reproducible handoff. State measured conditions and remaining targets explicitly.

Good client-facing case-study framing

A four-layer STM32 controller for industrial sensing and low-voltage actuation, with calibrated analog acquisition, protected outputs, isolated wired communication and hardware-enforced output inhibition. Once built, lead with the actual application outcome and measurements, then show the design decisions that made them possible.

What this review established

High confidence: supplied-sample inventory, literal topic counts with preserved source evidence, current planned feature set and missing native implementation. Medium confidence: how well the project will attract particular clients, because relevance depends on each job and on the quality of the completed board. Not established: electrical correctness, actual accuracy/current/temperature limits, DRC/ERC success or standards compliance.

No schematic/PCB analyzers, ERC/DRC, extracted-layout EMC/thermal analysis, whole-board SPICE or fabrication checks were run for this market review: the required native design does not yet exist. Existing calculations/simulation outputs were treated as preparation, not measured board validation. The earlier plant-monitor hardware was not supplied, so portfolio overlap is conditional.

Audit appendix: topic counts and cautions

Topic screen	Deduplicated briefs
STM32	26
ESP32	54
BLE / Bluetooth	45
Wi-Fi	25
Ethernet / PoE	17
RS-485	6
Modbus	3
CAN	12
CAN FD / FDCAN literal references	4
ADC	21
Analog / ADC / AFE / amplifiers	65
DAC / analog-output references	7
4-20 mA	0
0-10 V	2
Digital input / GPIO	16
Isolation / opto, any domain	10
Relay	14
PWM	6
Motor / servo / H-bridge / BLDC / stepper	43
Battery / Li-ion / LiPo / BMS	75
Charging / charger / BMS	41
Low power / sleep / power budgets	40
Four-layer	29
Firmware / embedded software	94
Test / validation / debug / prototype	185

Broad analog includes audio/AFE/amplifiers; the manually reviewed non-audio acquisition subset is 22 briefs, with five separate biomedical/force-sensing analog examples. CAN FD/FDCAN has four literal references but only two explicit FD-traffic examples. GPIO and general isolation counts do not establish industrial DI/isolation demand. Motor, charging and low-power screens include diverse applications; test/validation/debug/prototype is broader than bench testing.

Audit appendix: supporting delivery topics

Topic screen	Deduplicated briefs
Protection / ESD / surge / fuse	47
SI / impedance / differential pairs / returns	27
EMI / EMC	27
Manufacturing / fabrication / Gerber topics	162
BOM / bill of materials	114
SWD / JTAG / debug connector	8
Test points	12
I2C	23
SPI / QSPI	18
UART / USART	12
RTOS	5
EEPROM	2
Logging / logger / storage / timestamps	10
SD / microSD	7
USB	53
USB-C / Type-C	34
Buck / boost / DC-DC / LDO / supplies	76
Sensors / sensing / instrumentation	99
Mechanical / enclosure / mounting	119
DFM / DFA	40
DRC / ERC	25
Certification / standards topics	38
MQTT	5
OTA	9
Isolated power literal references	1
Gerber	77
Drill files	28
Pick-and-place / centroid / CPL	49
Editable / native design files	70
Bench testing / bring-up	53
Calibration	14

The companion dataset preserves titles, full descriptions, source locations, duplicate variants, snippets and manual notes. Counts are not weighted by budget or project size and do not predict hiring or earnings.

Sources, traceability and research limits

Supplied job files

Source	Attachment ID	Cards
F1	aae31026-0651-40ed-856d-0ecb146ca595	50
F2	a0f00578-d5fd-4f04-93c5-75917eeb29d1	50
F3	88ac0d1e-62e0-4f9c-9295-64ce372e32b1	50
F4	5bab1349-389d-4c71-9b41-cf93132f0115	50
F5	4a040272-0108-427a-9599-34ae7bd7c295	50
F6	2613b331-d561-40f3-a76b-ff213caf3f22	50
F7	fc55b17e-deb0-41c1-81ba-ed8f6e8413dd	50

Each source is the corresponding Pasted text.txt in C:/Users/huray/.codex/attachments/. Job IDs are local review identifiers, not Upwork job IDs. The companion index supplies original line locations. Full paths are preserved in unique_jobs.json.

Project evidence

Reviewed README.md, docs/Architecture.md, Interfaces.md, Engineering_Review.md, Design_Decisions.md, Schematic_Capture.md, Validation.md, circuit/component/calculation plans and hardware, firmware, mechanical and fabrication READMEs. project_scope.md preserves detailed file/line references. No existing project plan or design was changed.

Primary web sources checked October 4, 2026

S1: TI TIPD166 PLC analog-input reference design

<https://www.ti.com/tool/TIPD166>

Supports application coherence of industrial voltage/current acquisition, not Upwork demand frequency.

S2: ST STM32G474VE product/peripheral list

<https://www.st.com/en/microcontrollers-microprocessors/stm32g474ve.html>

Supports STM32 USB/FDCAN peripheral suitability; no native Ethernet MAC listed.

S3: Espressif ESP32-S3 TWAI documentation

<https://docs.espressif.com/projects/esp-idf/en/latest/esp32s3/api-reference/peripherals/twai.html>

ESP32-S3 native TWAI does not support CAN FD; swapping MCUs is not a drop-in way to preserve FD capability.

S4: Upwork portfolio guide

<https://www.upwork.com/resources/portfolio-guide>

Supports relevant personal project examples, clear outcomes and supporting media.

S5: Espressif ESP32-S3 product overview

<https://www.espressif.com/en/products/socs/esp32-s3>

Supports Wi-Fi/BLE capability for a potential later connected-controller extension.

Practical interpretation

Follow through with the controller if industrial/embedded work is the intended direction. Keep its functions cohesive, reduce completion risk, and publish measured proof. Treat wireless/battery experience as a portfolio-level coverage decision; do not force unrelated circuitry onto this board.